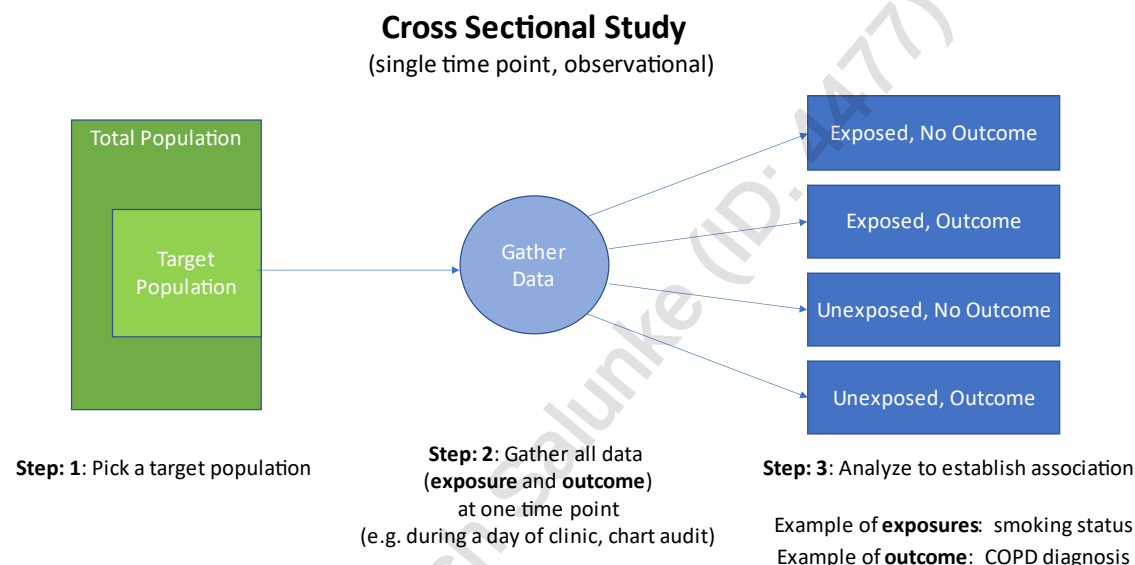


OBSERVATIONAL STUDY DESIGN: SUMMARY

Study Design	Summary	Advantages	Disadvantages	Further Reading
Case Report/Case Series	A report of one or several clinical events such as rare or unexpected disease findings or adverse events	• Cheap, quick and easy • Detects rare events	• Subject to selection, recall and reporting bias • No ability to examine prevalence or incidence. • No ability to infer causation	Riley DS et al. CARE guidelines for case reports: explanation and elaboration document. J Clin Epidemiol. 2017 May 18. pii: S0895-4356(17)30037-9. doi: 10.1016/j.jclinepi.2017.04.026.
Example:	Quick Example: A pharmacist notes a very strange reaction to a new antihypertensive a patient has started on and thinks that it may be worthwhile letting the greater medical community know. Additional Example: Title: An Adverse Drug Reaction to Trimethoprim-Sulfamethoxazole Revealing Primary HIV: A Case Report and Literature Review Summary: HIV-associated dermatologic reactions to antimicrobials, especially TMP-SMX, are more common in patients with lower CD4 counts. This is well defined in AIDS but has not been reported in primary HIV infection. A 55-year old male presented with shortness of breath and skin rash 3-days after incision and drainage of a gluteal abscess with prescription of TMP-SMX. He developed multiple genital lesions while in hospital and was found to have antibody-negative, HIV RNA + primary infection. Purported exposure was 15 days prior to presentation. Key Point: Hypersensitivity reactions in the setting of primary HIV infection may be possible and should be considered in the context of epidemiologic exposure risks. This may also help explain the mechanism of increased hypersensitivity reactions to TMP-SMX in patients with advanced HIV/AIDS. Most patients with primary HIV do not know they are infected, nor are they exposed to TMP-SMX. Therefore, other study designs would be unlikely to capture the rare population and rare exposure necessary to test a hypothesis. Reference: Meyer C et al. Case Reports in Infectious Diseases, Volume 2015, Article ID 691010, http://dx.doi.org/10.1155/2015/691010			
Cross Sectional	Study of a target population (i.e. hospital, health unit) at a <i>single point</i> in time to assess exposures, outcomes and related factors	• Cheap, quick, not so easy. • Ethically safe	• Cannot state incidence. • Cannot assign cause and effect relationships.	Wang X and Cheng Z. CHEST 2020;158(1): S65-71. DOI: https://doi.org/10.1016/j.chest.2020.03.012

Example:

Quick Example: A one-day audit of all patients in a health system asking about smoking (EXPOSURE) and diagnosis of COPD (OUTCOME).



Additional Example

Title: Point prevalence survey of antimicrobial use and healthcare-associated infections in Belgian acute care hospitals: results of the Global-PPS and ECDC-PPS 2017

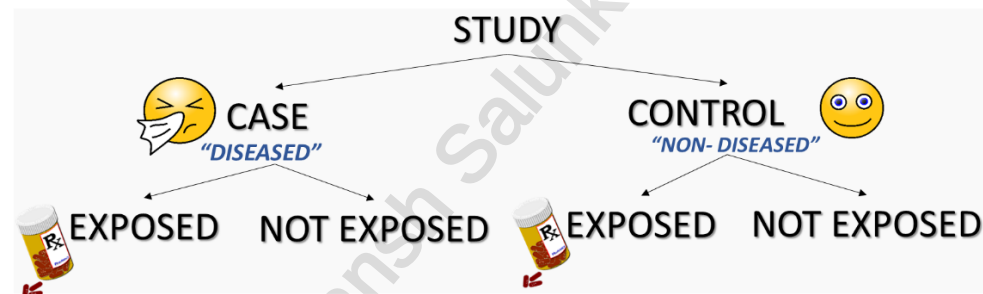
Summary: Understanding antimicrobial usage and infection in hospitals is key to identify areas of difference in prescribing and opportunities for antimicrobial stewardship interventions. Additionally, having a broad geographic catchment area assists in understanding intrahospital and/or regional differences in antimicrobial use. The Global Point Prevalence Survey (Global-PPS) and European Center for Disease Prevention and Control Point Prevalence Study (ECDC-PPS) are used commonly to collect, analyze and report this data. This cross-sectional study across Belgium's one-hundred and ten acute care hospitals found prevalence antimicrobial usage to be 27.1% (95% confidence interval 26.5-27.6%), guideline concordance of 76.6% of the time. Antibiotic stop dates were entered in only 40% of cases. Of all inpatients, there was a 7.3% (95% confidence interval 6.8-7.7%) prevalence of hospital-acquired infection.

Key Point: Across all sites, antimicrobial usage was frequent and guideline discordant in about one-quarter of cases. There is room to improve on both this metric as well as the stop date metric. Comparison to prior studies reveal a stable but not improving trend in use, concordance, and HAI prevalence. This cross-sectional study was done over a short period of time and captures a large amount of data with a relatively small outlay of human and capital resources. This study informs next steps for interventions in the areas studied.

Reference: Vandael et al. Antimicrobial Resistance and Infection Control (2020) 9:13 <https://doi.org/10.1186/s13756-019-0663-7>

Case-Control	Retrospective study that defines CASES (people with disease), matched to CONTROLS (people without disease) and then looks backwards to find EXPOSURES (risk factors for disease).	• Cheap, quick, fairly easy • Ethically safe • Can make a correlation between exposure and outcome	• Cannot prove a causal relationship between exposure and outcome • Dependent upon data quality • Subject to recall, interviewer, disclosure bias • Selecting matching control group not always simple	Mann CJ Observational research methods. Research design II: cohort, cross sectional, and case-control studies Emergency Medicine Journal 2003;20:54-60. http://dx.doi.org/10.1136/emj.20.1.54
---------------------	---	--	---	--

Example: **Quick Example:** A research wants to examine the relationship between lung cancer (CASE) and smoking history (EXPOSURE). The research looks at patients who have been diagnosed with lung cancer, a CONTROL group of people who have not been diagnosed with lung cancer but are otherwise similar and then asks patients about their smoking history (EXPOSURE).



Additional Example:

Title: Influenza vaccination, pneumococcal vaccination and risk of acute myocardial infarction: matched case-control study

Summary: Prior association between respiratory virus infection and acute myocardial infarction has been established. It is not known whether influenza or pneumococcal vaccination is protective. This study used data from electronic databases in the United Kingdom to match patients with myocardial infarction (CASES) and without myocardial infarction (CONTROLS) and then to evaluate the vaccination status (EXPOSURE) in each group. Using a 1(case):4(control) matching, 16012 cases and 62964 controls were included. Noted confounders for vaccination (conditions that would lead to vaccination AND myocardial infarction) were more common in the case group. Once adjusted for these factors, influenza vaccination was protective for MI with an odds ratio of 0.81 (95% confidence interval 0.77-0.85, $p < 0.001$). There was no significant association between pneumococcal vaccination and myocardial infarction with an odds ratio of 0.96 (95% confidence interval 0.91-1.02).

Key Point: As hypothesized, there appears to be a protective effect of influenza vaccination on the occurrence of MI. This association was not seen with pneumococcal vaccination. This study was able to rapidly (using a pre-collected dataset) identify people with and without disease and establish an association with prior exposure. Although we cannot say vaccination prevents MI, using prior study that suggests influenza can promote MI and that vaccination is associated with reduced MI risk, a compelling argument can be made for further investigation into the topic, either cohort studies or interventional studies.

Reference: A. Niroshan Siriwardena, Stella M. Gwini and Carol A.C. Coupland CMAJ October 19, 2010 182 (15) 1617-1623; DOI: <https://doi.org/10.1503/cmaj.091891>

Cohort Study	Prospective (or retrospective) study that identifies EXPOSURE and then looks forward (or backward) to find DISEASE.	• Can predict population prevalence of a condition. • Can have better data quality (prospective)	• Cannot prove causation (due to unmeasured confounders) but a more robust design than case-control. • More expensive and time consuming than case-control.	Barrett D, Noble HWhat are cohort studies?Evidence-Based Nursing 2019;22:95-96. doi: https://doi.org/10.1136/bmj.l5894
Example	Quick Example: A researcher wonders what impact daily exercise has on cardiac outcomes. They enroll people who exercise regularly (EXPOSED) and people who do not exercise regularly (UNEXPOSED) and follow them over time to evaluate their cardiac outcomes later in life (DISEASE). <pre> graph TD STUDY --> EXPOSED["EXPOSED (TO TREATMENT)"] STUDY --> NOT_EXPOSED["NOT EXPOSED (TO TREATMENT)"] EXPOSED --> EXPOSED_DISEASE["DISEASE (Sad Face)"] EXPOSED --> EXPOSED_NO_DISEASE["NO DISEASE (Happy Face)"] NOT_EXPOSED --> NOT_EXPOSED_DISEASE["DISEASE (Sad Face)"] NOT_EXPOSED --> NOT_EXPOSED_NO_DISEASE["NO DISEASE (Happy Face)"] </pre> Additional Example Title: Transmissibility and potential for disease progression of drug resistant <i>Mycobacterium tuberculosis</i>: prospective cohort study. Summary: <i>M. tuberculosis</i> is known to be most transmissible to close contacts in the same household or living environment. It is not known whether the antimicrobial resistance pattern plays a role in the transmissibility. This study was a prospective cohort conducted in Lima, Peru evaluation family member contacts of patients with multi-drug resistant TB(MDR TB) (EXPOSED group) versus family member contacts of those without MDR TB (NOT EXPOSED group). Family members were followed to evaluate new onset TB. The EXPOSED group had an 8% higher rate of infection (95% confidence interval 4-13%) compared to the NOT EXPOSED group over the follow up period. There was no difference in the development of active disease between groups. Key Point: In a large prospective cohort study, data collection was accurate enough to detect a difference in transmission risk between MDR and non-MDR TB to household contacts. This association would be difficult to support using case-control methodology owing to the follow up needs of the household contacts and not just patients. It may be possible to do this in a randomized controlled trial but it would be much more expensive, time consuming and ultimately not give significantly better information than this well conducted prospective cohort study. This data may move policy makers to identify MDR TB more rapidly to reduce transmission risks. Reference: Becerra M C, Huang C, Lecca L, Bayona J, Contreras C, Calderon R et al. Transmissibility and potential for disease progression of drug resistant <i>Mycobacterium tuberculosis</i>: prospective cohort study BMJ 2019; 367 :l5894 doi:10.1136/bmj.l5894			